Inheritance

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

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1 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: B

Explanation

A gene is a section of a molecule of DNA that codes for a specific protein. A describes something made of chromosomes, which is wrong because genes are located on chromosomes, not the other way round. C is a single base, which is far too small to code for a protein. D describes an enzyme, which is a protein that a gene may code for, not the gene itself.

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2 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: D

Explanation

Gametes such as egg cells are made by meiosis and are haploid, with 23 chromosomes in humans. Skin cells and white blood cells are body cells with the diploid number of 46. A zygote is formed when two haploid gametes fuse, so it is diploid.

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3 · Part 2

Paper 2 only | Inheritance | [1]

Answer: A

Worked solution

Work along the strand one base at a time: A→T, T→A, G→C, C→G, C→G, A→T. This gives TACGGT.

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4 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: C

Explanation

Heterozygous means having two different alleles of a gene, for example Bb. Homozygous means two identical alleles (BB or bb). Recessive describes an allele that is only expressed when two copies are present. Phenotype is the observable characteristic, not the allele combination.

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5 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: A

Explanation

Mitosis occurs during growth, repair, cloning and asexual reproduction, so repairing a cut in the skin involves mitosis. Sperm cells and haploid cells in an ovary are made by meiosis. Fusion of a sperm cell with an egg cell is fertilisation, which is not cell division.

6 · Part 2

Paper 2 only | Inheritance | [1]

Answer: B

Worked solution

Gametes from cow: R and R. Gametes from bull: R and W. Offspring: RR, RW, RR, RW. Two of the four are RR (red), so the expected percentage is $2 \div 4 \times 100 = 50\%$.

7 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: D

Explanation

A scar is caused by an injury, which is an environmental factor, so it is not inherited at all. Eye colour is controlled by genes. Height and body mass are affected by both genes and the environment, such as diet.

8 · Part 2

Paper 2 only | Inheritance | [1]

Answer: C

Explanation

Translation takes place at a ribosome, where the mRNA codons are read and amino acids are joined in order. The nucleus is where transcription (making mRNA) happens. Chromosomes carry the genes but are not the site of translation. The cell membrane controls what enters and leaves the cell.

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9 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: A

Explanation

Resistant bacteria are not killed by the antibiotic, so they keep reproducing and the infection continues. B is wrong because antibiotics do not turn bacteria into viruses. C is a common misconception: it is the bacteria that become resistant, not the patient. D is wrong because antibiotics do not stop white blood cells working.

10 · Part 2

Paper 1 & 2 | Inheritance | [1]

Answer: C

Worked solution

Mother XX gives X in every egg. Father XY gives X or Y in equal numbers. Offspring: XX, XY, XX, XY, so 2 out of 4 (0.5) are boys. This is the same for every pregnancy, whatever the sex of earlier children.

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11 · Part 2 Paper 1 & 2 Inheritance [5]

Mark scheme

(a) **[1]**

- the entire DNA of an organism (1)

ALLOW all the genetic material of an organism

(b) **[1]**

- nucleus (1)

(c) **[1]**

- 23 (1)

(d) **[2]**

- both cells were produced from the zygote OR fertilised egg by mitosis (1)
- mitosis produces cells with identical sets of chromosomes OR produces genetically identical cells (1)

Worked solution

(a) The genome is the entire DNA of an organism. A gene is just one section of that DNA.

(b) Chromosomes, which carry the genes, are found in the nucleus.

(c) Human gametes are haploid, so they contain 23 chromosomes (half of the diploid number, 46).

(d) Every body cell comes from the zygote by repeated mitosis. Mitosis produces cells with identical sets of chromosomes, so liver cells and skin cells contain the same genes. They look and work differently because they use different genes, not because they have different genes.

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12 · Part 2

Paper 2 only | Inheritance | [7]

Mark scheme

(a) **[3]**

- two strands (1)
- coiled to form a double helix (1)
- strands linked by paired bases, adenine with thymine and cytosine with guanine (1)

ALLOW A–T and C–G

(b) **[2]**

- thymine 27 (1)
- cytosine 23 AND guanine 23 (1)

(c) **[2]**

- RNA is single stranded (but DNA has two strands) (1)
- RNA contains uracil (U) instead of thymine (T) (1)

Worked solution

(a) DNA is made of two strands coiled together to form a double helix. The two strands are linked by pairs of bases: adenine (A) always pairs with thymine (T), and cytosine (C) always pairs with guanine (G).

(b) Every adenine pairs with a thymine, so thymine = 27%. Adenine and thymine together make 27 + 27 = 54%. The rest, 100 − 54 = 46%, is cytosine and guanine. These pair with each other, so each is 46 ÷ 2 = 23%. Check: 27 + 27 + 23 + 23 = 100%.

(c) An RNA molecule is single stranded, and it contains the base uracil (U) instead of thymine (T).

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13 · Part 2 Paper 2 only Inheritance [8]

Mark scheme

(a) **[3]**

- takes place in the nucleus (1)
- the two DNA strands separate OR the DNA unwinds (1)
- mRNA is made with bases complementary to the template strand (1)
- the mRNA leaves the nucleus (and goes to a ribosome) (1)
- max 3

(b) **[1]**

- AUG CCU AAA (1)

(c) **[1]**

- methionine, proline, lysine (in this order) (1)

ECF from (b)

(d) **[3]**

- tRNA carries a specific amino acid to the ribosome (1)
- each tRNA has an anticodon (1)
- the anticodon pairs with the complementary codon on the mRNA (1)
- so the amino acids are joined in the order set by the codons on the mRNA (1)
- max 3

Worked solution

(a) In the nucleus, the DNA of the gene unwinds and the two strands separate. One strand is used as a template, and an mRNA molecule is made with bases complementary to it. The mRNA then leaves the nucleus and travels to a ribosome.

(b) mRNA is complementary to the template strand, with uracil (U) pairing with adenine (A) instead of thymine: T→A, A→U, C→G, G→C. So TAC → AUG, GGA → CCU and TTT → AAA. The mRNA is AUG CCU AAA. Check: pairing back from AUG gives TAC, which matches.

(c) Look up each mRNA codon in the table: AUG = methionine, CCU = proline, AAA = lysine. (Reading the template codons TAC, GGA and TTT directly is a common mistake: the table entries UAC, GGA and UUU would give the wrong amino acids.)

(d) During translation, each tRNA molecule carries one specific amino acid to the ribosome. The tRNA has three bases called an anticodon, which pair with the complementary codon on the mRNA. This makes sure each amino acid is placed in the right position, so the amino acids are joined in the order coded for by the mRNA.

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14 · Part 2 Paper 1 & 2 Inheritance [5]

Mark scheme

(a) **[1]**

- an alternative form of a gene (1)

> **ALLOW** a different version of a gene

(b) **[1]**

- rr (1)

(c) **[1]**

- round seeds (1)

(d) **[2]**

- genotypes RR and Rr both give round seeds (1)
- because R is dominant, so one copy of R is enough for round seeds OR r is not expressed when R is present (1)

Worked solution

(a) An allele is an alternative form of a gene. Here the seed-shape gene has two alleles, R and r.

(b) Homozygous means two identical alleles, and recessive means the r allele, so the genotype is rr.

(c) Rr contains one dominant R allele, so the phenotype is round seeds.

(d) A plant with round seeds could be RR (homozygous) or Rr (heterozygous). Because R is dominant, a single copy of R gives round seeds, so the phenotype is the same even though the genotypes are different.

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15 · Part 2

Paper 2 only | Inheritance | [7]

Mark scheme

(a) **[1]**

- both alleles are expressed in the phenotype of a heterozygous organism (1)

ALLOW neither allele is dominant and both show in the phenotype

(b) **[2]**

- BB in one cell AND WW in one cell, placed correctly (1)
- BW in the other two cells (1)

ALLOW WB for BW

(c) **[2]**

- black, speckled and white phenotypes all named (1)
- in the ratio 1 black : 2 speckled : 1 white (1)

ECF from (b)

(d) **[2]**

- cross a black chicken with a white chicken OR cross BB with WW (1)
- every chick gets one B allele and one W allele, so all are BW (and speckled) (1)

Worked solution

(a) Codominance means that both alleles are expressed in the phenotype of a heterozygous organism. A BW chicken shows both black and white, so its feathers are speckled.

(b) Combine each gamete in the top row with each gamete in the first column:

	B	**W**
B	BB	BW
W	BW	WW

(c) BB = black, BW = speckled, WW = white. There is one BB, two BW and one WW, so the phenotype ratio is 1 black : 2 speckled : 1 white.

(d) A black chicken (BB) only makes B gametes and a white chicken (WW) only makes W gametes. Every chick receives one of each, so every chick is BW and has speckled feathers. Crossing two speckled chickens would not work, because it also gives black and white chicks.

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16 · Part 2 Paper 1 & 2 Inheritance [6]

Mark scheme

(a) **[2]**

- 10 + 3 = 13 students (1)
- (13 ÷ 40) × 100 = 32.5 (%) (1)

ALLOW 33 (%) OR 32 (%)

(b) **[2]**

- height is controlled by many genes (1)
- each gene has a small effect, so many different combinations of alleles give many different heights (1)

(c) **[2]**

- blood group falls into a small number of distinct groups OR a person belongs to one group or another (1)
- there are no intermediate blood groups, but height has a continuous range of values (1)

Worked solution

(a) Students with a height of 170 cm or more are in the 170–179 group (10 students) and the 180–189 group (3 students), so 13 students. Percentage = (13 ÷ 40) × 100 = 32.5%. Check: 32.5% of 40 = 0.325 × 40 = 13.

(b) Height is an example of polygenic inheritance: it is controlled by many genes rather than one. Each gene adds a small amount to height, so the many different combinations of alleles produce a wide range of heights with many in-between values.

(c) Blood group is controlled by a single gene, so each person falls into one of a few distinct groups and there are no in-between values. Height can be any value within a continuous range.

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17 · Part 2 Paper 1 & 2 Inheritance [6]

Mark scheme

(a) **[1]**

- N and n (1)

(b) **[2]**

- NN, Nn and nn (1)
- in the ratio 1 NN : 2 Nn : 1 nn (1)

(c) **[1]**

- 0.25 (1)

ALLOW 1 in 4 OR 1/4 OR 25%

(d) **[2]**

- 3/4 of the offspring have normal wings OR 320 × 0.75 (1)
- 240 (1)

ECF from (b)

Worked solution

(a) Each Nn parent passes on one allele in each gamete, so its gametes are N or n (in equal numbers).

(b) Combining N or n from one parent with N or n from the other gives NN, Nn, nN and nn. So the genotypes are NN : Nn : nn in the ratio 1 : 2 : 1.

(c) Only nn flies have vestigial wings: 1 out of 4, so the probability is 0.25.

(d) NN and Nn flies have normal wings: 3 out of 4. Expected number = 0.75 × 320 = 240. Check: vestigial = 0.25 × 320 = 80, and 240 + 80 = 320.

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18 · Part 2

Paper 1 & 2 Inheritance [7]

Mark scheme

(a) **[1]**

- BB AND Bb (1)

(b) **[4]**

- the brown rabbit is bb, so all its gametes carry b (1)
- if the black rabbit is BB, all offspring are Bb and have black fur (1)
- if the black rabbit is Bb, about half the offspring are expected to have brown fur OR a ratio of 1 black : 1 brown (1)
- so any offspring with brown fur shows the black rabbit is Bb (1)
- max 4

(c) **[2]**

- only six offspring is a small number (1)
- if the black rabbit is Bb, each offspring has a 0.5 probability of being black, so all six could be black by chance (1)

ALLOW the probability of six black offspring from Bb × bb is 1/64

Worked solution

(a) Black fur needs at least one dominant B allele, so the black rabbit is BB or Bb.

(b) A brown rabbit shows the recessive phenotype, so it must be bb and every gamete carries b. If the black rabbit is BB, all its gametes carry B, every offspring is Bb and all have black fur. If the black rabbit is Bb, half its gametes carry b, so about half the offspring would be bb with brown fur (1 black : 1 brown). Seeing any brown offspring proves the black rabbit is Bb.

(c) Six is a small sample. Even if the black rabbit were Bb, each offspring would have a 0.5 chance of being black, so six black offspring in a row could happen by chance (probability 0.5 × 0.5 × 0.5 × 0.5 × 0.5 × 0.5 = 1/64). More offspring would be needed to be confident.

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19 · Part 2 Paper 1 & 2 Inheritance [5]

Mark scheme

(a) **[2]**

- 309 ÷ 103 (1)
- 3 : 1 (1)

ALLOW 3.0 : 1

(b) **[2]**

- both parents are Pp (1)
- the green seedlings are pp, so each must have received a p allele from each parent, AND the parents have purple stems so they also have a P allele (1)

ALLOW a 3 : 1 ratio is expected from crossing Pp with Pp

(c) **[1]**

- 1/3 (1)

ALLOW 0.33 OR 1 in 3 OR 33%

Worked solution

(a) Divide the larger number by the smaller: 309 ÷ 103 = 3.0, so the ratio is 3 : 1. Check: 103 × 3 = 309.

(b) The green seedlings are pp, so each one received a p allele from both parents. Both parents have purple stems, so each also has a P allele. Both parents must be Pp. A Pp × Pp cross gives a 3 : 1 ratio of purple to green, which matches (a).

(c) The purple seedlings are PP, Pp and Pp in a 1 : 2 ratio of PP to Pp. Only 1 out of every 3 purple seedlings is PP, so the probability is 1/3. (Answering 1/4 is a common mistake: that is the fraction of all seedlings that are PP, not the fraction of purple seedlings.)

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20 · Part 2 Paper 1 & 2 Inheritance [6]

Mark scheme

(a) **[1]**

- ff (1)

(b) **[2]**

- the child (ff) has received one f allele from each parent (1)
- the parents do not have cystic fibrosis, so each must also have an F allele, so each is Ff (1)

(c) **[1]**

- 0.25 (1)

ALLOW 1/4 OR 1 in 4 OR 25%

(d) **[2]**

- probability of a girl = 0.5 (1)
- 0.25 × 0.5 = 0.125 (1)

ALLOW 1/8 OR 12.5%

ECF from (c)

Worked solution

(a) Cystic fibrosis is caused by a recessive allele, so the child must have two copies: ff.

(b) The child got one allele from each parent, so each parent must carry f. Neither parent has cystic fibrosis, so each must also have a dominant F allele. Both parents are Ff.

(c) Ff × Ff gives FF, Ff, fF and ff. Only ff has cystic fibrosis, so the probability is 1/4 = 0.25 for each child.

(d) The probability of a girl is 0.5, and this does not depend on the cystic fibrosis alleles. Probability of a girl with cystic fibrosis = 0.25 × 0.5 = 0.125 (1/8).

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21 · Part 2 Paper 1 & 2 Inheritance [5]

Mark scheme

(a) **[2]**

- body cells: mother XX AND father XY (1)
- gametes: mother X AND father X or Y (1)

(b) **[2]**

- all egg cells contain an X chromosome (1)
- sperm cells contain either X or Y; an X sperm gives a girl (XX) and a Y sperm gives a boy (XY) (1)

(c) **[1]**

- 50 (%) (1)

Worked solution

(a)

	Mother	Father
Sex chromosomes in a body cell	XX	XY
Sex chromosomes in a gamete	X	X or Y

(b) Every egg contains one X chromosome, so the egg cannot decide the sex. Half the sperm contain X and half contain Y. If an X sperm fertilises the egg, the child is XX (a girl); if a Y sperm fertilises it, the child is XY (a boy).

(c) The father's XY pair is separated when sperm are made, so half (50%) of sperm cells carry a Y chromosome.

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22 · Part 2 Paper 1 & 2 Inheritance [5]

Mark scheme

(a) **[2]**

- any two from: growth; repair; cloning; asexual reproduction (2)
- one correct (1)

(b) **[1]**

- 48 (1)

(c) **[2]**

- each chromosome is copied before the cell divides (1)
- each new cell receives an identical set of chromosomes OR one copy of every chromosome (1)

ALLOW DNA is copied for the first marking point

Worked solution

(a) Mitosis happens during growth, repair, cloning and asexual reproduction. Any two of these earn the marks.

(b) Mitosis produces cells with the same number of chromosomes as the parent cell, so each new cell has 48.

(c) Before the cell divides, each chromosome is copied. When the cell divides, each new cell receives one copy of every chromosome, so the two cells have identical sets of chromosomes and are genetically identical.

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23 · Part 2

Paper 1 & 2 | Inheritance | [6]

Mark scheme

(a) **[3]**

- four cells (1)
- each with half the number of chromosomes OR haploid (1)
- genetically different from each other (1)

(b) **[1]**

- 46 (1)

(c) **[2]**

- each gamete is genetically different (because of meiosis) (1)
- which sperm fertilises which egg is random, so each zygote gets a different combination of alleles (1)

Worked solution

(a) Meiosis produces four cells. Each has half the number of chromosomes of the original cell (they are haploid), and they are genetically different from each other.

(b) A human sperm and egg each have 23 chromosomes, so the zygote has 23 + 23 = 46, the diploid number.

(c) Every sperm and every egg is genetically different. Any sperm could fertilise any egg, and which ones fuse is a matter of chance. So each child receives a different combination of alleles from the parents, and brothers and sisters are genetically different.

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24 · Part 2 Paper 1 & 2 Inheritance [6]

Mark scheme

(a) **[1]**

- mitosis (1)

(b) **[2]**

- runners are produced by mitosis (from cells of the parent plant) (1)
- so the new plants are genetically identical to the parent OR are clones with the same alleles (1)

(c) **[1]**

- seeds give genetically varied plants, so some may have new useful features, for example resistance to a disease (1)

ALLOW seeds can be stored OR transported more easily than runners

(d) **[2]**

- the fields have different environmental conditions, for example less light, less water, fewer mineral ions OR a lower temperature in one field (1)
- the plants are genetically identical, so the difference is caused by the environment (1)

Worked solution

(a) Runners are a type of asexual reproduction, so their new cells are produced by mitosis.

(b) Mitosis produces cells with identical sets of chromosomes. The runner plants are clones of the parent plant, so they all have the same alleles, including the alleles for large, sweet fruit.

(c) Sexual reproduction produces genetic variation. Some seedlings might have a useful new combination of features, such as resistance to a disease, that the parent plant lacks.

(d) The two plants have the same genes, so any difference must be due to the environment. One field might give less light, less water, fewer mineral ions or lower temperatures, so the plant there grows less well and produces smaller fruit.

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25 · Part 2 Paper 1 & 2 Inheritance [7]

Mark scheme

(a) **[2]**

- (5.0 − 2.1) ÷ 2.1 × 100 (1)
- 138 (%) (1)

ALLOW 140 (%) OR 138.1 (%)

(b) **[3]**

- body mass (1)
- for identical twins (same genes), raising them apart increases the mean difference in body mass a lot, from 2.1 kg to 5.0 kg (1)
- but the mean difference in height only increases slightly, from 1.6 cm to 1.9 cm (1)

ALLOW use of the percentage increase from (a) for the second marking point

(c) **[2]**

- non-identical twins have different genes OR different alleles (1)
- both types of twin were raised together, so their environments were similar, so the extra difference is caused by genes (1)

Worked solution

(a) Change = 5.0 − 2.1 = 2.9 kg. Percentage increase = (2.9 ÷ 2.1) × 100 = 138% (to 3 significant figures). Check: 2.1 × 2.38 = 5.0.

(b) Identical twins have the same genes, so any difference between them is caused by the environment. When identical twins are raised apart, the mean difference in body mass rises from 2.1 kg to 5.0 kg, more than double. The mean difference in height only rises from 1.6 cm to 1.9 cm. So body mass is affected more by the environment.

(c) Both groups were raised together, so their environments were similar. The non-identical twins have different alleles, while identical twins have the same genes. So the larger difference in height (4.6 cm compared with 1.6 cm) is mainly caused by genetic differences.

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26 · Part 2 Paper 2 only Inheritance [7]

Mark scheme

(a) **[2]**

- a rare, random change in genetic material OR DNA (1)
- that can be inherited (1)

(b) **[2]**

- X-rays (ionising radiation) increase the incidence of mutations (1)
- the lead apron absorbs X-rays, so it reduces the staff's exposure and the chance of mutations in their cells (1)

ALLOW staff are exposed to X-rays many times, so protection is needed

(c) **[2]**

- as the time exposed increases, the number of mutant colonies increases (1)
- any correct quantitative statement, for example the number increases by about 8 for every extra 10 s OR it increases steadily from 2 to 33 OR the increase is roughly linear (1)

REJECT directly proportional (there are 2 mutant colonies at 0 s)

(d) **[1]**

- mutations happen naturally OR randomly, even without exposure to ultraviolet rays (1)

Worked solution

(a) A mutation is a rare, random change in genetic material that can be inherited.

(b) Ionising radiation such as X-rays increases the incidence of mutations. Staff take X-ray images many times each day. The lead apron absorbs the X-rays, so less radiation reaches their body cells and the chance of mutations is reduced.

(c) The longer the bacteria are exposed to ultraviolet rays, the more mutant colonies there are. The increase is steady: about 7 to 9 more colonies for every extra 10 s, from 2 colonies at 0 s to 33 colonies at 40 s.

(d) Mutations are random and happen naturally at a low rate, so a few mutant colonies appear even without ultraviolet rays. Ultraviolet rays only increase how often mutations happen.

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27 · Part 2

Paper 2 only | Inheritance | [7]

Mark scheme

(a) **[1]**

- proline, glutamic acid, glutamic acid (1)

(b) **[1]**

- the second amino acid changes from glutamic acid to valine (1)

(c) **[3]**

- the protein folds into a different shape (1)
- so the protein does not work properly OR its function changes (1)
- this changes a characteristic of the organism OR changes the phenotype, for example haemoglobin carries oxygen less well (1)

(d) **[2]**

- the same amino acid (glutamic acid) is coded for, so the sequence of amino acids does not change (1)
- so the protein has the same shape and works in the same way (1)

Worked solution

(a) From the table: CCU = proline, GAG = glutamic acid, GAG = glutamic acid.

(b) The second codon becomes GUG, which codes for valine. So the second amino acid changes from glutamic acid to valine; the other two are unchanged.

(c) The sequence of amino acids decides how a protein folds and what shape it has. A different amino acid can change the shape of the protein, so it may not work properly. In this example the haemoglobin works less well, which affects the person's phenotype.

(d) GAA and GAG both code for glutamic acid, so the amino acid sequence stays the same. The protein has the same shape and works normally, so the phenotype does not change. This is one reason why most mutations have no effect on the phenotype.

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28 · Part 2 Paper 2 only Inheritance [6]

Mark scheme

(a) **[2]**

- (8 ÷ 500) × 100 (1)
- 1.6 (%) (1)

(b) **[2]**

- the number of mutations increases (1)
- because chemical mutagens increase the incidence of mutations (1)

(c) **[2]**

- a mutation in a cell that produces gametes can be present in a gamete, which passes it on at fertilisation (so it is in every cell of the offspring) (1)
- a skin cell does not produce gametes, so the mutation is not passed to the offspring (1)

Worked solution

(a) Percentage = (8 ÷ 500) × 100 = 1.6%. Check: 1.6% of 500 = 0.016 × 500 = 8. The data show that most mutations have no effect on the phenotype, some have a small effect and only rarely do they have a significant effect.

(b) A chemical mutagen increases the incidence of mutations, so more new mutations would be expected in the exposed zebrafish.

(c) Only genetic material in gametes is passed to the next generation. If a cell that makes gametes has a mutation, a gamete can carry it, and after fertilisation the mutation is in every cell of the offspring. A skin cell does not make gametes, so a mutation there stays in that person's skin and is not inherited.

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29 · Part 2 Paper 1 & 2 Inheritance [7]

Mark scheme

(a) **[5]**

- there is variation in fur colour in the population (1)
- the variation is caused by different alleles OR genes (that may arise by mutation) (1)
- mice with light fur are easier for owls to see, so are more likely to be eaten (1)
- mice with dark fur are more likely to survive (1)
- and reproduce (1)
- passing on the alleles for dark fur to their offspring (1)
- over many generations the proportion of mice with dark fur increases (1)
- max 5

(b) **[2]**

- (76 − 20) ÷ 30 OR 56 ÷ 30 (1)
- 1.9 (% per year) (1)

ALLOW 1.87 (% per year)

Worked solution

(a) The mice vary in fur colour, and this variation is caused by different alleles. On dark soil, light mice are easier for owls to see, so more of them are caught and eaten. Dark mice are more likely to survive and reproduce. They pass on the alleles for dark fur to their offspring. Over many generations, the proportion of mice with dark fur increases until most of the population has dark fur.

(b) Change in percentage = 76 − 20 = 56. Rate = 56 ÷ 30 = 1.87, or 1.9% per year (2 significant figures). Check: 1.87 × 30 = 56.

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30 · Part 2

Paper 1 & 2 Inheritance [8]

Mark scheme

(a) **[2]**

- the percentage of infections caused by resistant bacteria increases (over the whole period) (1)
- any correct quantitative statement, for example from 8% in 2000 to 34% in 2015 OR an increase of 26 percentage points OR the largest increase was between 2005 and 2010 (1)

(b) **[4]**

- a mutation gave some bacteria resistance to the antibiotic (1)
- the antibiotic kills the bacteria that are not resistant (1)
- resistant bacteria survive (1)
- and reproduce, passing on the gene OR allele for resistance (1)
- so the proportion of resistant bacteria in the population increases (1)
- max 4

(c) **[2]**

- any two from (2):
- only prescribe the antibiotic when it is needed OR not for viral infections
- make sure patients complete the full course of the antibiotic
- use a different antibiotic OR change antibiotics regularly
- good hygiene, for example hand washing, to stop resistant bacteria spreading
- one correct (1)

Worked solution

(a) The percentage rises in every five-year period, from 8% in 2000 to 34% in 2015, an increase of 26 percentage points. The biggest rise (12 percentage points) was between 2005 and 2010.

(b) A random mutation gave a few bacteria resistance to the antibiotic. When the antibiotic is used, it kills the bacteria that are not resistant, but the resistant ones survive. They reproduce and pass on the gene for resistance, so over many generations the resistant bacteria make up a larger proportion of the population. This makes infections harder to control.

(c) Using the antibiotic less often reduces the advantage that resistant bacteria have. The hospital could prescribe it only when it is really needed (never for viral infections), make sure patients finish the full course, change to different antibiotics, and use good hygiene so that resistant bacteria do not spread between patients.

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